Introduction to Machine Learning

Lecture 3 — Supervised Learning

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Today's agenda

1. Recap
2. What is a model?
3. Types of learning
4. Supervised Learning
5. The learning loop
6. Linear regression
7. Decision trees
8. Overfitting
9. Remedies
10. Lab preview

Recap

What is a model?

> A model is a function that maps inputs to outputs — learned from data rather than hand-coded.
> — Bishop, *Pattern Recognition and Machine Learning*

Types of learning

Supervised — labelled training data

- Classification: "Is this email spam?"
- Regression: "What will the price be?"

Unsupervised — no labels

- Clustering: "Which customers are similar?"
- Dimensionality reduction: "What's the structure?"

Supervised Learning

The learning loop

```
graph LR
    A[Training data] --> B[Model]
    B --> C[Prediction]
    C --> D{Loss function}
    D -->|Gradient| B
    D --> E[Repeat until convergence]
```

Linear regression — intuition

We want to find the line of best fit through our data points.

Formally, we learn parameters **w** and **b** such that:

$$\hat{y} = w \cdot x + b$$

minimises the **mean squared error** over all training examples:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Linear regression — in Python

```
from sklearn.linear_model import LinearRegression
import numpy as np

X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2.1, 3.9, 6.2, 7.8, 10.1])

model = LinearRegression()
model.fit(X, y)

print(f"Slope:     {model.coef_[0]:.2f}")
print(f"Intercept: {model.intercept_:.2f}")
print(f"Predicted for x=6: {model.predict([[6]])[0]:.2f}")
```

Decision trees — intuition

A decision tree partitions the feature space using a series of binary questions. Each internal node is a feature split; each leaf is a prediction.

```
graph TD
    A{Age > 30?} -->|Yes| B{Income > 50k?}
    A -->|No| C[Predict: No]
    B -->|Yes| D[Predict: Yes]
    B -->|No| E[Predict: No]
```

Comparing models

	Linear regression	Decision tree
Output type	Continuous	Continuous or categorical
Interpretable	✅ Easy	✅ Easy
Non-linear relationships	❌	✅
Sensitive to outliers	✅ Yes	Less so
Prone to overfitting	Rarely	Very much so

Overfitting

What is overfitting?

> A model overfits when it learns the training data so well that it fails to generalise to new examples.

The model has memorised noise, not signal.

Bias–variance tradeoff

High bias (underfitting) ——————————————————— 20%

Just right ——————————————————————————————— 80%

High variance (overfitting) ——————————————————————— 100%

Remedies for overfitting

Decision trees

- Limit max depth (`max_depth=4`)
- Require minimum samples per leaf
- Prune after training

General techniques

- More training data
- Cross-validation
- Regularisation (L1 / L2)
- Ensemble methods (next week)

Quick check

Which model would you use for predicting house prices?

Respond in the poll

Lab preview

In today's lab you will:

- Load the Boston Housing dataset
- Fit a linear regression model and a decision tree
- Evaluate both on a held-out test set
- Plot the bias–variance curves

> ⚠️ **Warning: Deadline**
> Submit your notebook to Gradescope by **Friday 23:59**.

Lab environment walkthrough

Смотреть видео: Lab environment walkthrough

Resources

- Course notes: Chapter 3 — Linear Models
- Scikit-learn docs: `sklearn.linear_model`, `sklearn.tree`

Bishop, C. (2006). Pattern Recognition and Machine Learning. Springer, Ch. 3.1–3.3.

See you Thursday.